Domain model (1/2): stays, rooms, guests and money

Money (integer millimes) is the type of every amount.
build_bill() turns a Reservation, its charges, discounts, applicable taxes and payments into a Bill; an Invoice freezes its lines.
Ids (guest_id, room_id…) are the associations drawn as lines.

Rooms and prices

«entity»
HotelSettings

check_in_time: time
check_out_time: time
cancellation_fee_bp: int
cancellation_window_hours: int
grace_period_hours: int
discount_cap_bp: int
weekend_surcharge_bp: int
weekend_nights: str
anonymise_after_months: int
home_country: str
/weekend_days: frozenset[int]

+validated(): HotelSettings
+cancellation_policy(): CancellationPolicy
+to_strings(): dict[str, str]
{static} +from_strings(stored): HotelSettings

«create»

«value object»
CancellationPolicy

fee_basis_points: int
window: timedelta
check_in_time: time
grace_period: timedelta

+arrival_instant(arrival, tz): datetime
+free_until(reservation, tz): datetime
+fee_for(reservation, now, tz): Money
+no_show_fee(reservation, now, tz): Money

«value object»
Pricing

periods: tuple[RatePeriod, …]
weekend_nights: frozenset[int]
weekend_surcharge_bp: int

+rate_for(room, night): Money

«entity»
RatePeriod

id: int | None
category_id: int
name: str
period: StayPeriod
nightly_rate: Money
category_name: str
created_at: datetime | None
/last_night: date

«entity»
RoomCategory

id: int | None
name: str
description: str
version: int
room_count: int

Reservations

«entity»
Reservation

id: int | None
code: str
guest_id: int
period: StayPeriod
status: ReservationStatus
rooms: list[BookedRoom]
nights: list[RoomNight]
cancellation_fee: Money | None
notes: str
created_by: int | None
created_at: datetime | None
cancelled_at: datetime | None
checked_in_at: datetime | None
checked_out_at: datetime | None
no_show_at: datetime | None
version: int
guest_name: str
/accommodation_total: Money
/current_rooms: list[BookedRoom]
/occupants: int

+nights_of(room_id): list[RoomNight]
+ensure_can_become(to): None
+transition(to): None
+check_in(now, today): None
+check_out(now, today): bool
+move_room(from_room_id, to_room_id, today, number, category_name, rate_for): list[date]

«entity»
BookedRoom

room_id: int
adults: int
children: int
number: str
category_name: str
moved_out_on: date | None
/is_current: bool
/occupants: int

«entity»
RoomNight

room_id: int
night: date
rate: Money

«enumeration»
ReservationStatus

CONFIRMED
CHECKED_IN
CHECKED_OUT
CANCELLED
NO_SHOW

«value object»
StayPeriod

arrival: date
departure: date
/nights: int

+overlaps(other): bool
+contains(day): bool
+each_night(): Iterator[date]

«entity»
MaintenancePeriod

id: int | None
room_id: int
period: StayPeriod
reason: str
created_by: int | None
created_at: datetime | None
/first_night: date
/last_night: date
/reopens_on: date

«entity»
Room

id: int | None
number: str
category_id: int
capacity: int
nightly_rate: Money
in_service: bool
notes: str
version: int
category_name: str | None
housekeeping: Housekeeping
archived_at: datetime | None
/is_archived: bool

+can_host(occupants): bool

«enumeration»
Housekeeping

DIRTY
CLEAN
INSPECTED

Bills and payments

«entity»
Invoice

id: int | None
number: str
reservation_id: int
reservation_code: str
guest_name: str
arrival: date
departure: date
issued_at: datetime
issued_by_name: str
lines: list[BillLine]
payments: list[Payment]
listed_total: Money | None
/bill: Bill
/total: Money
/balance: Money

«value object»
Money

millimes: int

{static} +zero(): Money
{static} +from_dinars(dinars): Money
{static} +parse(text): Money
+times(quantity): Money
+percent(basis_points): Money
+is_zero(): bool
+is_negative(): bool
+to_decimal(): Decimal

«value object»
Bill

lines: list[BillLine]
payments: list[Payment]
/accommodation: Money
/extras: Money
/fees: Money
/discount: Money
/subtotal: Money
/tax: Money
/total: Money
/paid: Money
/balance: Money

«entity»
FixedDiscount

value: Money

+amount(accommodation): Money
+validate(): None

«entity»
PercentDiscount

rate_bp: int

+amount(accommodation): Money
+validate(): None

«entity»
Discount

{abstract}

id: int | None
reservation_id: int
reason: str
approved_by: int | None
approved_by_name: str
created_at: datetime | None
removed_at: datetime | None
/is_active: bool

+amount(accommodation): Money
+validate(): None

«entity»
Payment

id: int | None
reservation_id: int
kind: PaymentKind
method: PaymentMethod
amount: Money
paid_at: datetime
reference: str
recorded_by: int | None
recorded_by_name: str
/signed_amount: Money

«value object»
BillLine

kind: LineKind
label: str
quantity: int
unit: Money
amount: Money
first: date | None
last: date | None
rate_bp: int | None

«enumeration»
PaymentKind

DEPOSIT
PAYMENT
REFUND

«enumeration»
PaymentMethod

CASH
CARD
TRANSFER

«enumeration»
LineKind

NIGHTS
EXTRA
FEE
DISCOUNT
DISCOUNT_CAP
TAX

Extra services

«entity»
ServiceCharge

id: int | None
reservation_id: int
extra_id: int
description: str
quantity: int
unit_price: Money
charge_date: date
created_by: int | None
created_at: datetime | None
voided_at: datetime | None
voided_by: int | None
void_reason: str | None
/amount: Money
/is_void: bool

«entity»
Extra

id: int | None
name: str
unit_price: Money
is_active: bool
version: int
archived_at: datetime | None
/is_archived: bool

Guests

«entity»
Guest

id: int | None
first_name: str
last_name: str
id_doc_type: IdDocType
id_doc_number: str
nationality: str | None
phone: str | None
email: str | None
notes: str
is_vip: bool
do_not_rent: bool
do_not_rent_reason: str | None
is_active: bool
created_at: datetime | None
version: int
anonymised_at: datetime | None
archived_at: datetime | None
/is_archived: bool
/is_anonymised: bool
/full_name: str
/search_name: str
/sort_name: str
/phone_digits: str | None
/masked_doc_number: str

+anonymise(now): None
+apply(details): list[str]

«enumeration»
IdDocType

CIN
PASSPORT

Taxes (Strategy pattern)

«entity»
TaxExemption

id: int | None
reservation_id: int
tax_rule_id: int
reason: str
created_at: datetime | None
created_by_name: str
tax_name: str

+validate(): None

«entity»
PercentTax

rate_bp: int
/kind: TaxKind

+amount(stay): Money
+basis(stay): tuple[int, Money]
+validate(): None

«entity»
PerAdultNightTax

amount_per: Money
/kind: TaxKind

+amount(stay): Money
+basis(stay): tuple[int, Money]
+validate(): None

«entity»
TaxRule

{abstract}

id: int | None
name: str
applies_to: TaxBase
payers: TaxPayers
is_active: bool
version: int
archived_at: datetime | None
/is_archived: bool
/kind: TaxKind

+concerns(national): bool
+amount(stay): Money
+basis(stay): tuple[int, Money]
+validate(): None

«enumeration»
TaxBase

ACCOMMODATION
SERVICES

«enumeration»
TaxPayers

EVERYONE
FOREIGN
NATIONALS

«enumeration»
TaxKind

PERCENT
PER_ADULT_NIGHT

«value object»
TaxableStay

accommodation: Money
services: Money
adult_nights: int